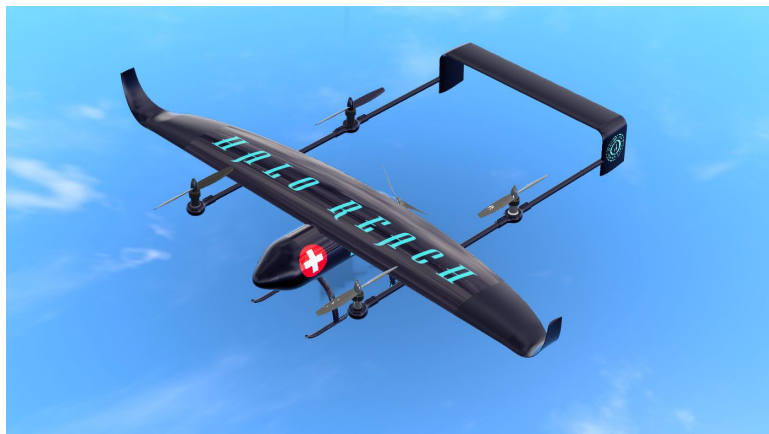

HALO REACH: Wing Loads and Structural Design

UAS for Disaster Relief - AE3 2024



Student: Stephen Liu

CID: 02015258

Supervisor: Dr Siti Ros Shamsuddin

Department: Department of Aeronautics

Course: MEng Aeronautical Engineering

Module: AERO60004 - Group Design Project

Date: 17/06/2024

Academic Year: 2023-2024

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.



Table 1. People involved in HALO REACH - GDP (**student in bold**)

Team code	Team	Names	CID
FATT01	Team Leads	Moreno, Patricia	01857449
		Rashid, Hasan H	02020313
FATT02	Aerodynamics	Alam, Shahriar	02037395
		Dirirsa, Naboth	02026211
		Lawlor, Max F	02026199
		Singh, Parasdeep	02099678
		Homa, Matthew M	02038403
FATT03	Structures	Leng, Jiaxun (Aaron)	02015060
		Wang, Zhiyan	01911209
		Liu, Stephen	02015258
		Chauhan, Shivani	02096006
FATT04	Systems and propulsion	Lin, Yongkai	02029838
		Chursuk, Tharm	02048996
		Wankhade, Divij P	02027906
		Aggarwal, Ishaan K	02030969
		Halim, George	02059571
FATT05	Flight dynamics	Baikady, Thilak S	02016794
		Scott-Brown, Ruairidh	01734080
		Allen, Harry	02028116
		Tang, Shun S	02023280
		Lee, Jeongheon (Edward)	01754474
FATT06	Add-ons	Rosenthal, Adam T	02027024
		Ramteke, Shreya	02079050
		Nakhla, Andrew	02044646

Abstract

A twin-boom fixed-wing VTOL (Vertical Take-Off and Landing) aircraft is selected to blend the aerodynamic efficiency of fixed-wing aircraft in cruise with VTOL ability of rotorcraft. This presents a challenging structural problem for the wing, which will have to sustain the boom, rotor and tail loads. Due to significant payload and system weights, a mass constraint of 7.5 kg per wing is imposed. High specific-mass properties are desired, so an all-composite wing is proposed. A cantilever box-beam wing with two spars, three ribs and skin-stringer panels is subsequently designed to ensure the wing can survive the loads experienced in takeoff, cruise and landing at the ultimate load factor. Spars and ribs were designed using Composite Laminate Theory to prevent buckling and strength failure. In the absence of suitable composite analyses techniques for D-section and skin-stringer panels, quasi-isotropic layups were used to allow for isotropic techniques and datasheets to be applied for design purposes. This yields an initial mass of 7.47 kg per wing, within the design constraint. FEA (Finite Element Analysis) is finally used to verify the design of each component and to suggest modelling improvements for the next design iteration.

Contents

Table of Contents	ii
List of Figures	iv
List of Tables	vi
Nomenclature	vii
1 Introduction	1
2 Wing Planform Geometry	1
3 Wing Loads	1
3.1 Wing Idealisation	2
3.2 Load Approximations	2
3.3 Load Analysis	2
4 Wing Structural Component Sizing	3
4.1 Spar Placement	3
4.2 Wing Spar Design	5
4.3 D-Section Design	6
4.4 Skin-Stringer-Ribs Configuration	7
4.4.1 Method	7
4.4.2 Initial Design	8
4.4.3 Final Design	9
4.5 Summary	9
5 Computational Verification	10
5.1 FEA Model Setup	10
5.2 FEA Results	10
6 Discussion and Next Steps	11
References	13
A Load Diagrams	15
B Load Analysis Subcases	15
C Derivation	17
D FEA	17

D.1	Spars	17
D.2	D-Section	18
D.3	Skin-Stringer Panels	19
D.4	Skin Panels	19
D.5	Ribs	20

List of Figures

1	Wing planform geometry	1
2	Beam external load diagram for cruise	1
3	External force diagram	3
4	Shear force diagram	3
5	Bending moment diagram	3
6	Torque diagram	3
7	Wingbox idealisation	4
8	Wingbox shear force diagram	4
9	Optimal Spar Locations	5
10	Shear flow spanwise distribution	5
11	Spar location as % chord	5
12	Wingbox Dimensions	5
13	Spar layups	6
14	Spar failure index, no ply drops	6
15	D section failure index	7
16	Z-stringer	7
17	Skin stringer initial design	9
18	Skin stringer final design	9
19	Wing structural design	10
20	Skin stringer panel at 1.15 - 2.3 m FEA setup, yellow is stringers, blue is skin	11
21	I-beam spar, first buckling mode	11
22	Skin stringer panel at 0-1.15 m, first buckling mode	12
23	Example simply supported skin-stringer panel, first buckling mode	12
24	Skin panel at 2.3-2.7 m, first buckling mode	12
25	Beam load diagram for takeoff	15
26	Beam load diagram for landing	15
27	Takeoff loading subcases	15
28	Cruise loading subcases	16
29	Landing loading subcases	16
30	Spar direct buckling model setup	17
31	Spar shear buckling model setup	17
32	Spar direct buckling mesh convergence study	18
33	Spar shear buckling mesh convergence study	18
34	First buckling mode of spar under direct load on flange	18
35	First buckling mode of spar under shear load on web	18
36	D-Section FEA model setup	18

37	D-section mesh convergence study	18
38	D-section first shear buckling mode	19
39	Skin panel at 2.3-2.7 m mesh convergence study	19
40	Skin panel at 2.7-3.0 m mesh convergence study	19
41	Rib at root FEA model	20
42	Rib at root mesh convergence	20
43	Rib at root, first buckling mode	20

List of Tables

1	People involved in HALO REACH - GDP (student in bold)	1
2	Web and flange layups	6
3	Parameter design space	8
4	Stringer geometry	8
5	Mass summary	10
6	FEA buckling comparison with analytical	11

Nomenclature

Abbreviations

ESDU	Engineering Science Data Unit
FEA	Finite Element Analysis
HALO REACH	Highly Aerodynamic Light Operation Remote Emergency Aid & Crisis Help
NACA	National Advisory Committee for Aeronautics
OEI	One Engine Inoperative
UAS	Uncrewed Aerial System
VTOL	Vertical Take-Off and Landing

Symbols

$\frac{\partial \theta}{\partial z}$	Twist rate along span (rad/m)
ν	Poisson's Ratio
τ	Shear Stress (Pa)
A	Area (m ²)
b	Width (m) or Stringer Pitch (m)
d	Stringer Flange Length (m)
E	Young's Modulus (Pa)
F	Wing Crushing Force (N)
f	Farrar Efficiency
G	Shear Modulus (Pa)
h	Height (m)
I	Moment of Inertia (N/m ⁴)
K	Buckling Coefficient
L	Rib Spacing (m)
M	Reaction Moments (Nm) or Bending Moment (Nm)
N	Load per unit length (N/m)
n	Number
q	Shear Flow (N/m)
R	Reaction Forces (N)
t	Thickness (m)
A	Composite A matrix
D	Composite D matrix

Subscripts

$B1$	Rear Boom
$B2$	Front Boom
box	Wingbox
c	Critical Buckling Load
D	D-section
eff	Effective
fl	Flange
i	ith wing span station
n	nth cell
s	Stringer
W	Wing
w	Web

1 Introduction

A 160 kg Vertical Take-Off and Landing (VTOL) Uncrewed Aerial System (UAS) is proposed that can perform two main missions: rapid supply relief delivery to disaster zones and a long-endurance communications relay operation. To achieve this, a twin-boom fixed wing VTOL aircraft configuration is selected to blend the aerodynamic efficiency of fixed-wing aircraft in cruise with the VTOL capability of rotorcraft [1]. The twin-boom configuration allows for suitable placement of rotors and the pusher propeller, but presents a challenging structural problem for the wing, which must carry the loads generated by the booms, rotor and tail. Due to large payload and system component masses, significant weight constraints are imposed on the structure, with a design constraint of 7.5 kg per wing [2]. High specific-mass properties are therefore desired, so an all-composite wing is proposed.

There are two widely adopted wing structures for monoplanes: strut-braced and cantilever. The former is more structurally efficient and can minimise wing mass, but the parasitic drag generated by the struts makes this configuration undesirable for endurance missions. Hence, a cantilever wing is preferred. Meanwhile, the internal wing structure can be broadly classified into two types: multi-spar and box beam structures. The former is the preferred lightweight solution to low aspect-ratio wings whereas the latter is preferred for high aspect-ratio wings [3]. The latter is selected as a high aspect ratio wing is proposed [4]. The box beam structure typically consists of two spars to effectively resist torsional shear and bending stresses, while the skin resists the bending loads. A skin-stringer panel configuration is often used to minimise skin mass without compromising structural integrity. Ribs are placed primarily to support the wing shape against crushing loads, but can also act as attachment points for control mechanisms and stringers.

2 Wing Planform Geometry

The wing planform in Fig. 1 shows a high aspect ratio wing (with aspect ratio of 8.5) mounted in a high-wing configuration, with a straight leading edge and piecewise tapered trailing edge [4]. The chord varies from 0.91 m at the root to 0.18 m at the tip. NACA 6409, a 9% thickness airfoil, is selected, leading to a maximum root thickness of 8.1 cm and tip thickness of 1.6 cm [4]. The booms are placed at 1.15 m from the fuselage centreline [5]. The area from 2.75 m outwards is a winglet mounted at 30° to the wing plane [6]. Fig. 1 shows the winglet projection onto the planform. There are no fuel or battery cells in the wing due to stability considerations [2, 5]. No high lift devices are used, with only ailerons to be placed on the wing [7].

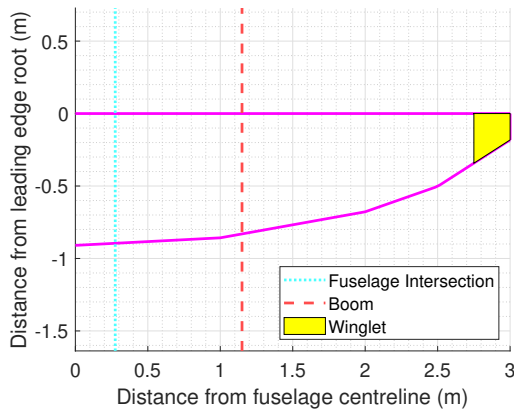


Figure 1. Wing planform geometry

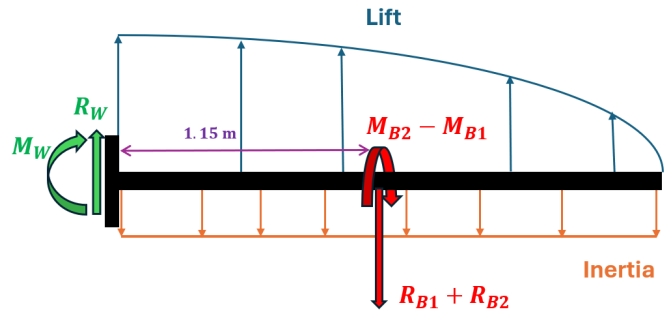


Figure 2. Beam external load diagram for cruise

3 Wing Loads

Load analysis is conducted for three main flight modes: takeoff in rotorcraft mode, cruise in fixed-wing mode, and landing in rotorcraft mode. For the supply delivery mission, the UAS must operate

with and without a 50 kg payload and 6.5 kg of fuel [8]. One engine inoperative (OEI) cases, where one rotor is inoperative, must be considered for takeoff and landing; no OEI case exists for cruise as there is no redundancy for the pusher propeller. Meanwhile, for the endurance mission, there is a fixed 20 kg payload, so mass variation is due to burning of the 36.5 kg fuel [8]. For each mission, different subcases involving a permutation of mission type, fuel mass, payload mass and OEI configuration are examined at the manoeuvre speed and dive speed [9]. The ultimate load factors for cruise are 3.75 and -1.5; for takeoff and landing the ultimate load factors are 5.25 and -0.75 [9].

3.1 Wing Idealisation

The wing is idealised as a cantilever beam, with an example beam load diagram seen in Fig. 2. Load diagrams for takeoff and landing can be found in Appendix A. Note that only the horizontal projection of the winglet is considered. The winglet is treated as part of the wing, and its aerodynamic effect is accounted for with a 2.3% increase in bending moment compared to the same wing planform with no winglet [6]. Since the booms are each modelled as a cantilever beam [10], the front and rear of each boom will transmit reaction loads to the wing. The contribution is noted in Fig. 2 as $R_{B1} + R_{B2}$ for boom reaction forces and $M_{B2} - M_{B1}$ for boom reaction moments, where subscript $B1$ denotes rear boom and $B2$ denotes front boom. Care is taken to ensure this is consistent with notation used for boom load diagrams [10]. Boom torque contributions are also considered, but are omitted from Fig. 2 for clarity. The wing loads are then transmitted via reaction force R_W , moment M_W and torque (omitted from Fig. 2 for clarity) to the fuselage.

3.2 Load Approximations

To generate the internal load diagrams, external forces acting on the wing must be characterised. Each wing is assumed to be 7.5 kg, and the inertial load is assumed to vary with the area (chord squared). For the cruise configurations, the lift across both wings is assumed to have a magnitude equal to the weight, scaled by the relevant load factors. The lift follows an elliptical distribution [4]. Since a high wing configuration is selected, the top of the wing is unaffected by fuselage intersection, so the entire planform can generate lift. Meanwhile, in the takeoff and landing configurations, there is a maximum drag of 4 N opposing the direction of motion due to the wing acting as a bluff body [7]. Takeoff and landing drag can therefore be neglected. Boom contributions are calculated separately [10]. All aerodynamic loads are assumed to act at the quarter-chord, and the shear centre is assumed to be at the half-chord for purposes of torque analysis. Boom contributions and inertial loads are assumed to act at the shear centre. The load distribution is seen in Fig. 3. Note that the spike at 1.15 m is due to rotor forces being transmitted to the wing during takeoff and landing [9].

3.3 Load Analysis

Loads can be discretised across discrete wing span stations, and the shear force on each station is computed by summing the forces acting on each span station between the wing tip and the element. By taking moments about the wing tip, the bending moment can be computed by considering the cumulative moment generated by the shear forces at all preceding span stations. The torque distribution, defined as positive nose-up, is the sum of the products of each characterised force and its distance from the shear centre. This yields the shear force, bending moment and torque diagrams seen in Fig. 4, Fig. 5 and Fig. 6. Negative shear force is plotted in Fig. 4 to allow easy comparison with Fig. 3 and Fig. 5. Only the maximum and minimum cases for takeoff, cruise and landing are presented for clarity; for a detailed breakdown of the subcases considered in load analysis, see Appendix B. Sharp change in internal forces at 1.15 m are due to rotor loads being transmitted to the wing.

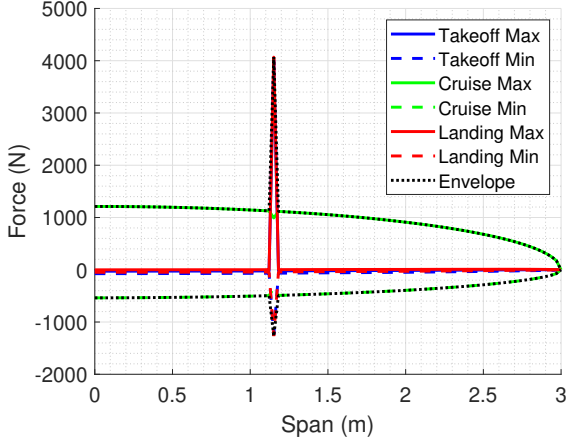


Figure 3. External force diagram

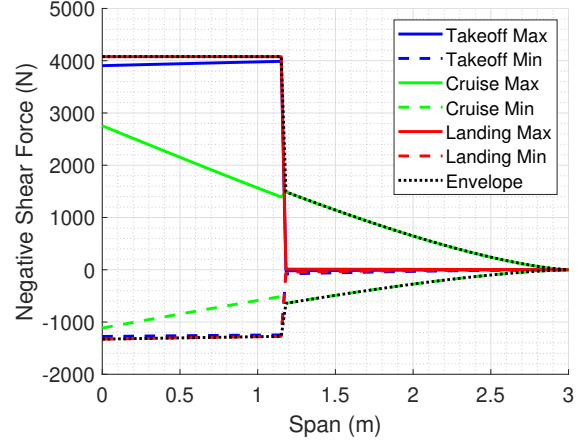


Figure 4. Shear force diagram

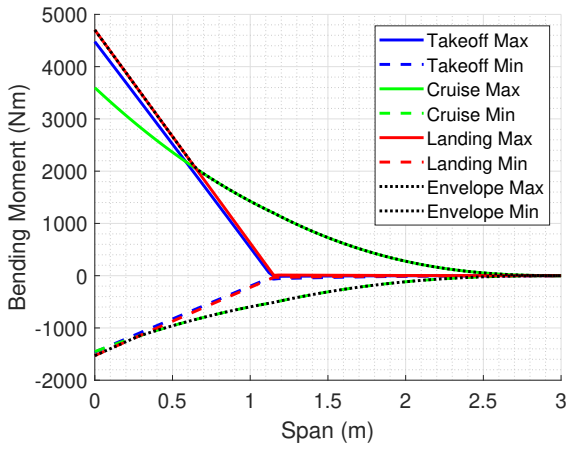


Figure 5. Bending moment diagram

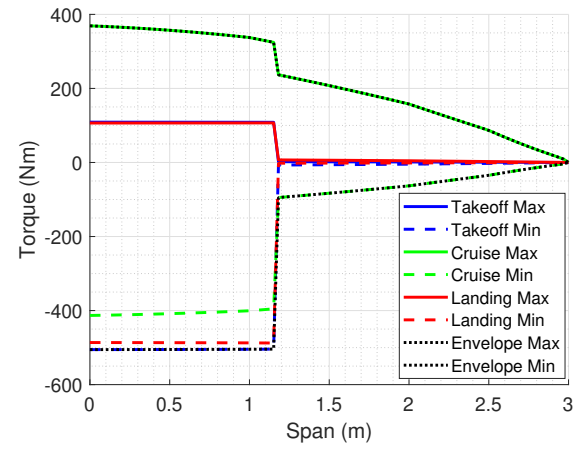


Figure 6. Torque diagram

4 Wing Structural Component Sizing

Three major components are designed: spars, skin-stringer panels and ribs. All components use a Toray 3960 Prepreg System unidirectional layup [10, 11]. A set of design principles are first established. Orthotropic layups are desired as symmetric, balanced layups lead to mechanically reliable layups. Only 0° , $\pm 45^\circ$ and 90° plies are considered, and there must be no more than 70% of any one ply orientation and no less than 10% of any ply orientation [12]. No more than two consecutive ply orientations further aids the design of a balanced laminate [12]. Moreover, quasi-isotropic laminates are desired to reduce risk of delamination as this configuration minimises out-of-plane interlaminar stresses. Therefore, a minimum of 8 plies are required and is used as a starting point for minimum mass design. Once layups are designed, the composite **ABD** matrices, the laminate constitutive equation, can be derived and analytical techniques employed to design components [13]. FEA (finite element analysis) is performed in Section 5 to verify the proposed designs.

4.1 Spar Placement

Spars must first be placed, after which the wingbox dimensions become fixed, allowing for analyses of all structural components. Front and rear spars are usually located at a constant % chord of the wing and run parallel to the leading and trailing edges respectively. The front spar can be placed at 25% chord to coincide with the lift force, but once aeroelastic effects are accounted for the recommended placement becomes 12-17 % chord; the back spar is usually placed at 60% chord to accommodate 30% chord ailerons [3]. However, following these guidelines leads to a rear spar with similar shape to the trailing edge seen in Fig. 1, which is significantly weaker than a straight spar.

Since two straight spars are desirable for structural integrity, spars will inevitably be placed at a variable % chord location, making the constant % chord placement guidelines difficult to implement. To aid spar placement, the spars' primary role in resisting shear is utilised. If the shear flow q acting across the spar can be minimised, then the relation $q = \tau t$, where τ is shear stress and t is thickness, suggests that thickness is minimised, leading to mass minimisation.

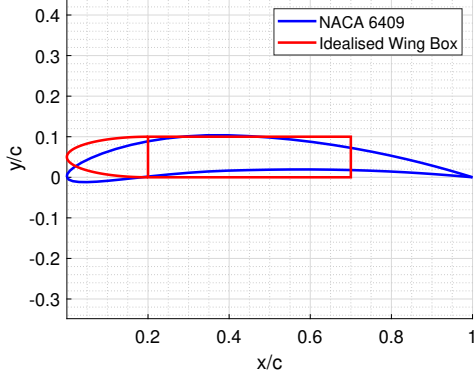


Figure 7. Wingbox idealisation

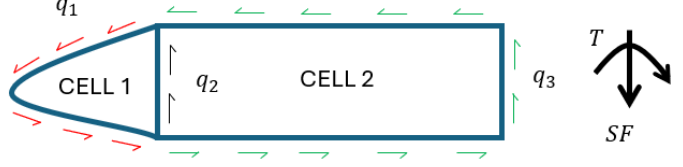


Figure 8. Wingbox shear force diagram

The wingbox is idealised as a rectangular box and a D-section (due to the absence of trailing edge devices), as seen in Fig. 7. The torque and shear force will generate shear flow across the multi-cell wingbox as seen in Fig. 8. It is assumed that shear force acts from the shear centre, now located at the midpoint between the spars. D-section contribution to shear centre location is neglected. The shear centre location here can therefore be different to the one used for load analysis in Section 3.2. To reconcile the difference, the shear force is shifted to the new shear centre from the half chord, and a suitable moment couple is added onto the torque to compensate for this shift. The shear flow can then be calculated by generating a set of simultaneous equations from Eq. (1), Eq. (2) and Eq. (3).

$$T = \sum_n 2A_n q_n \quad (1) \quad SF = h_{\text{box}}(q_1 - q_2 - q_3) \quad (2) \quad \frac{\partial \theta_n}{\partial z_n} = \frac{1}{2A_n G} \left(\oint \frac{q}{t} ds \right) \quad (3)$$

where A is cell area, T is torque, SF is shear force, G is shear modulus and h_{box} is wingbox height, taken as average airfoil thickness at front and rear spar locations. Eq. (1) is the Bredt-Batho equation, Eq. (2) describes force equivalence and Eq. (3) is for compatibility. For spar placement purposes, constant thickness is assumed. q_1 , q_2 and q_3 is solved to obtain shear flow across D-section, front spar and rear spar respectively. An optimisation problem can now be defined to place the spars:

$$\min \sum_{i=1}^{\text{sections}} |q_{1,i}| + |q_{2,i}| + |q_{3,i}| \quad (4)$$

By varying the front and rear spar start points at the root and endpoints at the tip, an optimisation algorithm, implemented via MATLAB's *fmincon* function, is run. Eq. (1) - (3) are solved at each spanwise station before the objective function in Eq. (4) is evaluated. The optimum spar placement is seen in Fig. 9 and the spanwise shear flow distribution is seen in Fig. 10. A distribution contextualising the spar locations in terms of chordwise position is shown in Fig. 11, and shows that the front spar is at 14% chord for most of the wing while the rear spar ranges from 55-70% chord for the majority of the wing, which lines up with general spar placement guidelines. More importantly, the spacing from the boom to 90% of the span, as seen in Fig. 9, is able to accommodate 30% chord ailerons, as seen in Fig. 11. This yields an aileron area of 0.32 m², which was deemed satisfactory by the Flight Dynamics team for lateral control [5]. With the spar placement set, the wingbox dimensions across the span can be computed, as seen in Fig. 12.

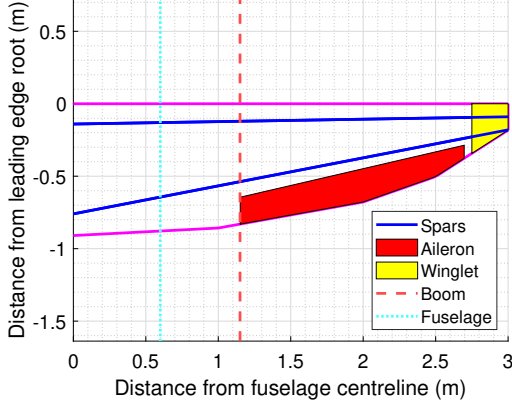


Figure 9. Optimal Spar Locations

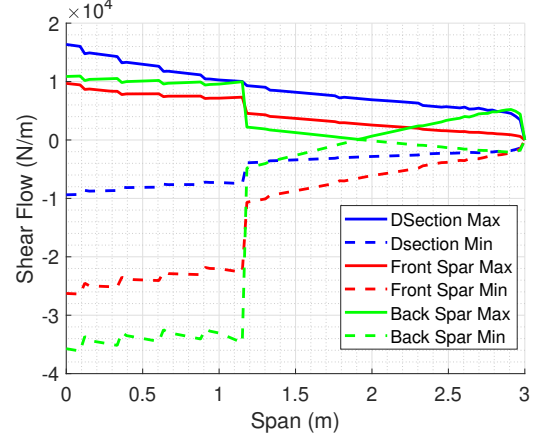


Figure 10. Shear flow spanwise distribution

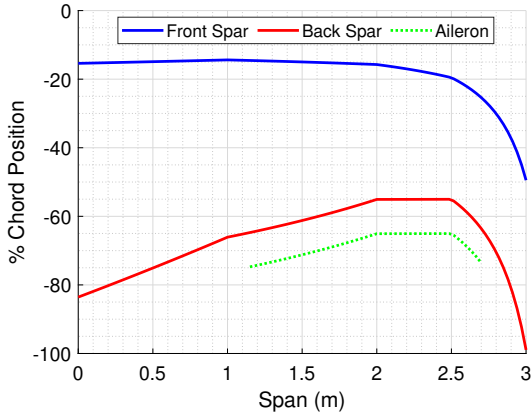


Figure 11. Spar location as % chord

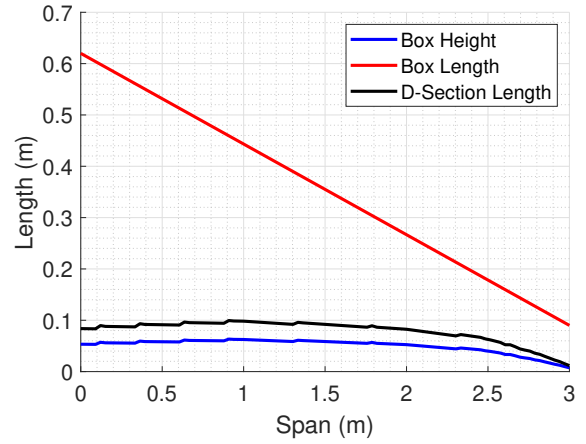


Figure 12. Wingbox Dimensions

4.2 Wing Spar Design

An I beam spar is selected due to its high bending resistance. For the first iteration, it is assumed that the web resists only shear and the flange resists only bending. Another consideration for the design process is manufacturing: a co-curing rather than co-bonding method is used to minimise loss of structural integrity in manufacturing. This requires web and flange layups to be cured together. To facilitate manufacturing, layups must be designed such that the web layup form the outer layers of the flange layup, as seen in Fig. 13. The web and flange must therefore be simultaneously sized.

An iterative design process is followed to size the spar. The webs are sized first and must resist the maximum absolute shear flow seen in Fig. 10. The web height h_w is set equal to the box height. The web must resist shear buckling, and by assuming the web is a long, rectangular flat plate with simply supported boundary conditions, the web critical shear buckling load can be computed as

$$N_{cw} = \frac{\sqrt{D_{w11}D_{w22}}}{h_w^2} \left(35 + 21.8 \frac{D_{w12} + 2D_{w33}}{\sqrt{D_{w11}D_{w22}}} \right) \quad (5)$$

where N_c is the critical buckling load per unit length, D_{ij} is the (i, j) entry in the composite $\mathbf{D}$ matrix [14]. Subscript w denotes web. The Tsai-Wu failure criterion is used to verify laminate strength [15, 16]. Once a suitable web layup has been designed, the flange layup can be designed, with web layup acting as the outer layers of the flange. By assuming the flange is a long plate with one simply supported and one free end, the critical flange buckling load can be found using:

$$N_{cfl} = \frac{12D_{33fl}}{b_{fl}^2} \quad (6)$$

where subscript fl denotes the flange and b_{fl} is the flange width [14]. b_{fl} is a design variable and can be selected such that the buckle score, a ratio of applied load to critical buckling load, equals 1 for maximum structural efficiency. By utilising beam theory to compute the compressive load acting on the flange, as well as the moment of inertia of the I-beam, the following equation can be derived

$$N_{fl} = \frac{3Mt_{fl}}{h_w(h_w t_w + 6b_{fl}t_{fl})} \quad (7)$$

where t is obtained upon designing the flange and web layups. See Appendix C for derivation. b_{fl} can now be sized by setting $N_{fl} = N_{cfl}$, which is solved via simultaneous equations of Eq. (6) and Eq. (7). Web and flange layups are iteratively designed until a minimum mass design is found.

Table 2. Web and flange layups

	Layup	Thickness (mm)	Length (mm)
Web	$[-45^\circ/45^\circ/0^\circ/90^\circ]_S$	1.5	Varies with box height
Flange	$[-45^\circ/45^\circ/0^\circ/90^\circ/0^\circ/90^\circ]_S$	2.2	19.2

The design is summarised in Table 2 and can be visualised using Fig. 13. The flange width is $b_{fl} = 19.2$ mm. Both the front and back spar utilise the same design, as they experience similar maximum shear flow. The spar height is assumed to equal the box height across the span, and flange width unchanged with span. The spar design is then verified across the span, as seen in Fig. 14. The design fails if either buckle score or Tsai-Wu index exceeds 1 at any point along the span. It seems the web is currently over-designed, with low buckling and failure indices. However, due to the outlined design methodology, an 8 ply minimum is imposed, so thinner layups cannot be pursued. Ply drops were experimented with for the flange, but the need to maintain quasi-isotropic properties to mitigate delamination made dropping the inner plies undesirable. The current design leads to an estimated mass of 0.49 kg per spar.

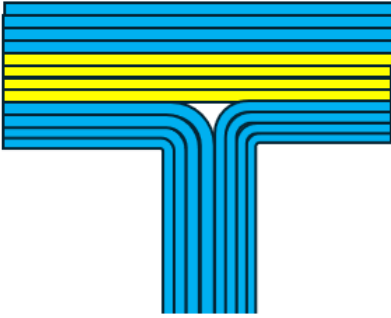


Figure 13. Spar layups

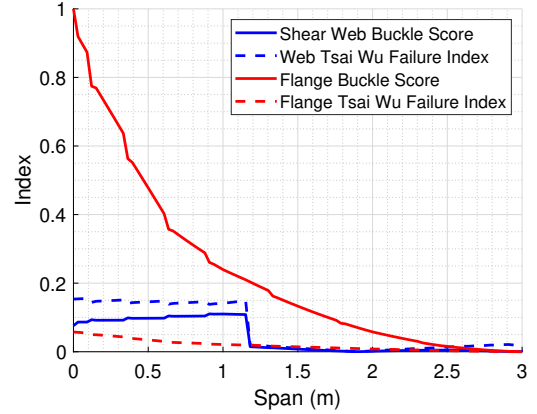


Figure 14. Spar failure index, no ply drops

4.3 D-Section Design

The D-section is a curved section that must resist the shear flow seen in Fig.10. Datasets for shear buckling of curved composite plates are unavailable, but datasets based on isotropic homogeneous plates are available. These isotropic datasets can be used with caution to size the composite D-section provided a quasi-isotropic composite layup is designed. The laminate can then be treated as a quasi-isotropic homogeneous material with equivalent properties. Equivalent properties, such as Young's modulus in the two principal directions (E_1 and E_2), shear modulus (G_{12}) and Poisson's ratio (ν_{12}) can be derived from the inverse of the composite $\mathbf{A}$ matrix using [17]:

$$E_1 = \frac{1}{A_{11}^{-1}t} \quad (8) \quad E_2 = \frac{1}{A_{22}^{-1}t} \quad (9) \quad G_{12} = -\frac{1}{A_{33}^{-1}t} \quad (10) \quad \nu_{12} = -\frac{A_{12}^{-1}}{A_{11}^{-1}} \quad (11)$$

Once equivalent laminate properties are derived, the critical shear buckling load of the quasi-isotropic plate is governed by the isotropic, homogeneous shear buckling equation:

$$N_c = KE_1 \frac{t^3}{b^2} \quad (12)$$

where $b = b_D$ is the circumferential length of the D-section panel and K can be obtained from ESDU 02.03.18 [18]. The D-section has a high aspect ratio, so the curve corresponding to infinity in ESDU 02.03.18 is used. An iterative process is conducted to determine the layup required for minimum mass, resulting in an 8 ply layup of $[-45^\circ/45^\circ/0^\circ/90^\circ]_S$ and a panel thickness of 1.5 mm. By using this ply layup throughout the span, the shear buckle score and failure indices across the span can be seen in Fig. 15, which suggests the layup is structurally safe. The buckle score does seem low and will be verified computationally in Section 5. The D-section has a mass of 0.44 kg.

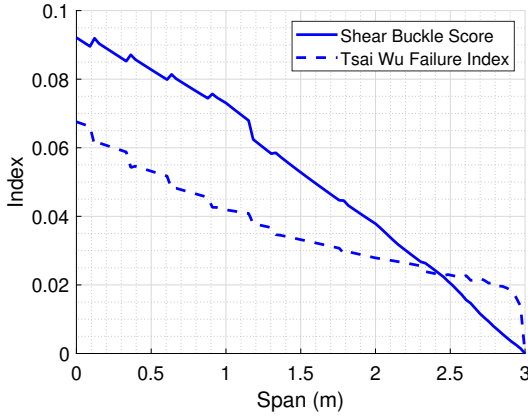


Figure 15. D section failure index

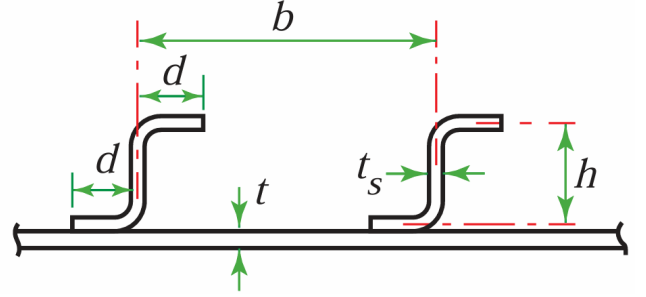


Figure 16. Z-stringer

4.4 Skin-Stringer-Ribs Configuration

Stringer reinforced skin panels is the standard procedure used to minimise the mass while ensuring the panels can resist compressive stress due to the bending moment. A Z-stringer configuration, seen in Fig. 16, is preferred as it provides enhanced structural advantages over unflanged stringers [3].

4.4.1 Method

There are no datasets detailing composite Z-stringer skin stringer panels. Stringers could be treated as simply-supported boundary conditions, allowing composite analysis on the thinner plates. However, this fails to capture the skin-stringer interaction and can lead to too conservative estimates. Meanwhile, FEA is too computationally expensive and mesh-sensitive to be suitable for preliminary design [19]. Hence, the quasi-isotropic method discussed in Section 4.3 is used here. The critical buckling load of the skin-stringer panel can be found with Eq. (12), where K is a buckling coefficient obtained from ESDU 71014 [20], b is stringer pitch and $t = t_{eff}$ is panel effective thickness. By assuming the stringers are smeared over the panel, t_{eff} can be computed using:

$$t_{eff} = t + \frac{A_s}{b} \quad (13) \quad A_s = t_s(h + 2d) \quad (14)$$

where A_s is the stringer area, t_s is the stringer thickness, h is stringer height, d is flange length and t is skin thickness. By considering a force-moment couple, the compressive load per unit length on the skin due to bending moment M can be derived by dividing M by $h_{box}b_{box}$, where b_{box} is wingbox width. Techniques used to design isotropic homogeneous skin-stringer panels can now be applied to the quasi-isotropic composite laminate. To preserve the quasi-isotropic assumption, skin and stringer must use the same layup to have same properties. Hence the $t_s/t = 1$ curve in ESDU 71014 is used.

Table 3. Parameter design space

Design Parameters	Parameter Space
n_s	5 - 30
h/b	0.1 - 1.0
d/h	0.3, 0.4, 0.5

Skin-stringer geometry can now be sized and the parameter space specified in Table 3 is explored. The number of stringers n_s is used as a parameter rather than b to ensure a whole number of stringers is always obtained. Meanwhile, h/b and d/h are constrained by the datasets provided by ESDU 71014. Once the layup and stringer geometry is obtained and shown to avoid strength and buckling failure, the Farrar efficiency, describing the structural efficiency in terms of initial and flexural buckling [3], can be obtained for each stringer configuration. Rib placement is then considered. Rib placement is coupled with skin-stringer design as thinner panels require more ribs to support the wing shape, leading to a tradeoff. The optimal rib spacing (L) to resist the wing crushing force can be found using

$$f = \frac{1}{t_{eff}} \sqrt{\frac{NL}{E_2}} \quad (15)$$

where f is the Farrar efficiency. Farrar efficiency should exceed 0.7 for an efficient structure [21]. The crushing force F that the ribs must support is given by

$$F = \frac{M^2 h_{box} L t_{eff} b_{box}}{2EI^2} \quad (16)$$

where I is the moment of inertia of both skin panels [21]. Rib placement must also provide attachment points for ailerons. Ribs are then sized by considering a low aspect ratio plate. Buckling can be considered using the simply supported composite plate buckling equation

$$N_c = \frac{1}{b^2} \left(K \sqrt{D_{11}D_{22}} + 2\pi^2 D_{12} + 4\pi^2 D_{33} \right) \quad (17)$$

where $b = b_{box}$ and K depends on aspect ratio and is found using ESDU 80023 [22].

4.4.2 Initial Design

Table 4. Stringer geometry

n_s	b (mm)	d (mm)	h (mm)	t (mm)	t_s (mm)
20	29.5	3.5	8.9	1.5	1.5

After following an iterative process, a $[0^\circ/90^\circ/-45^\circ/45^\circ]_S$ layup was utilised for both top and bottom skin panels. The stringer parameters can be found in Table 4. Note that the flange width is very small and will require an automatic tape layup machine with ± 0.1 mm tolerance to ensure structural integrity. This corresponds to $f = 0.7$ and an optimal $L = 1.15$ m at the root. All stringer geometry parameters are fixed across the span, with the stringer height corresponding to 20% of the box height for the majority of the span. The initial design is structurally safe over the entire span, as seen in Fig. 17. The quasi-isotropic method is compared against treating the stringers as simply-supported boundary conditions, and the higher composite buckle score in Fig. 17 compared to quasi-isotropic buckle score suggest that the quasi-isotropic method is valid as it follows the same trends and is less conservative than pure composite analysis. Farrar efficiency and buckling is only calculated up to 2.7 m as after this point the stringers exceed half the height of the wingbox, violating geometrical constraints. Fig. 17 show low buckling scores, implying low structural efficiencies. This will be refined in Section 4.4.3. It is also noted that other design configurations with higher buckling

scores and same mass as Table 4 had large stringer heights that protruded excessively into the wingbox. For example, one configuration that yielded the same mass as the initial design had a buckling score of 0.90, but the stringers for the top and bottom panel combined took up an unreasonable 98% of the wingbox height. The current selected stringer configuration is the highest stringer height deemed geometrically acceptable that also yields a minimum mass of 3.9 kg per panel.

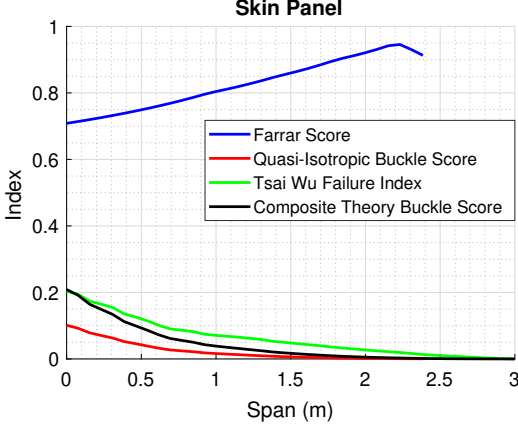


Figure 17. Skin stringer initial design

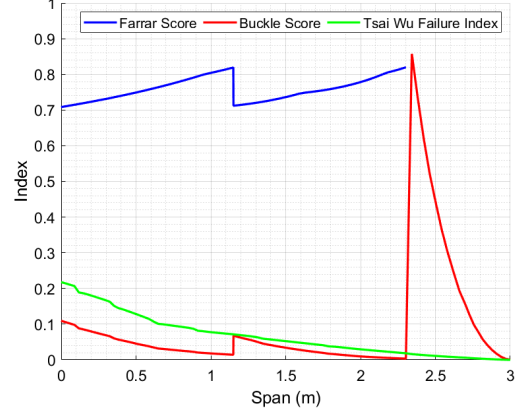


Figure 18. Skin stringer final design

An optimal rib placement of 1.15 m does not completely account for the need to place ribs in locations to facilitate control mechanism attachment. Moreover, the bending moment and the compression force decreases across the span as seen in Fig. 5, so by Eq. (15), the optimal rib spacing increases along the span. The first two ribs to be placed are at the root and at 1.15 m, which coincides with the boom. The rib can therefore serve as an attachment point for the boom and helps with the jump in internal loads experienced at that point. At 1.15m, the optimal spacing now exceeds the wing semi-span. Henceforth, ribs are placed to facilitate attachment points, leading to a rib placed at 2.7 m (end of the aileron). Ribs can now be sized, with each rib only requiring a layup of $[90^\circ / -45^\circ / 45^\circ / 0^\circ]_S$ to withstand the wing crushing force of 8,900 N. This yields a total rib mass of 0.20 kg.

4.4.3 Final Design

Now that ribs are placed, the skin stringer panel design is revisited to further optimise mass. A minimum 8 ply layup is used, which does not facilitate weight savings via ply drops. Instead, stringers can be terminated when no longer needed to save weight, provided $f > 0.7$. Stringers must not terminate in free space as the load they carry must be re-distributed to other components without inducing failure due to high stress concentrations; termination at ribs is the preference [3]. Considering stringer termination requirements, the stringer runout between the ribs at root and boom remain set by the initial design iteration. The runout between the ribs at boom and 2.7 m are decreased from 20 until a minimum number can be found such that $f > 0.7$ and no buckling occurs. This number was found to be 10, and carrying 10 stringers rather than 20 stringers from the boom onwards led to a mass decrease of 0.6 kg, reducing each panel mass to 3.3 kg, a 15% decrease. Stringer geometry, except pitch, remains unchanged. Due to low loads near the tip, it was hypothesised that stringers would be redundant after a certain point, with the skin alone sufficient to resist the loads. The location where stringers are no longer needed is 2.3 m, and removal of stringers at this point leads to a buckle score of close to 1, as seen in Fig. 18. To facilitate the stringer termination, an extra rib is placed at 2.3 m. This adds an extra 0.05 kg of rib mass, but leads to a further 0.4 kg skin panel mass reduction, with each panel now weighing 2.9 kg, a 26% decrease from the initial design.

4.5 Summary

The final wing structural configuration which resists strength and buckling failure can be visualised in Fig. 19. The total mass of each structural component is summarised in Table 5. Each wing has a structural mass of 7.47 kg, which sits within the 7.5 kg per wing design constraint.

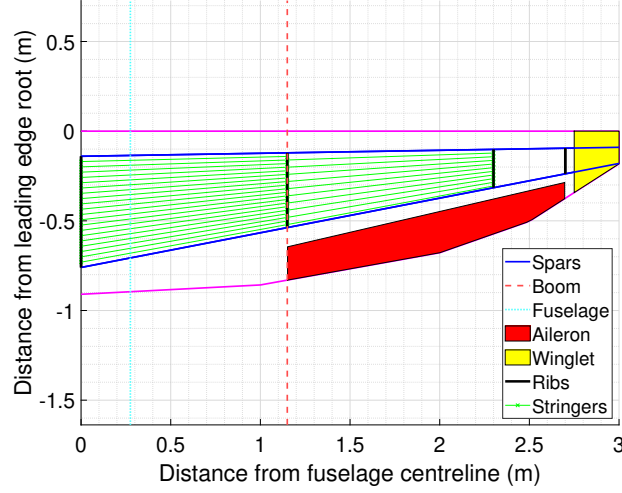


Figure 19. Wing structural design

Table 5. Mass summary

2xSpars (kg)	D-Section (kg)	2xSkin-Stringer Panels (kg)	4xRibs (kg)	Total (kg)
0.98	0.44	5.80	0.25	7.47

5 Computational Verification

Each component is modelled and verified in Abaqus, with a focus on buckling performance. If each individual component is successfully validated, a full wing model FEA can then be performed. If components fail, reasons for failure are determined and will be considered in the next design iteration.

5.1 FEA Model Setup

A summary of FEA model setup is provided here, and more details, including mesh convergence studies, can be found in Appendix D. Each component is modelled according to the dimensions and composite layup provided in Section 4. For each FEA model, the part is modelled as a shell and meshed with S4R shell elements. For the spars, the root has an encastre boundary condition imposed, while no constraints are imposed at the tip. Shell edge normal loads are utilised for direct buckling analysis, while shell edge shear loads are used for shear buckling analysis. Note for direct buckling only one flange is loaded while for shear buckling only the web is loaded; this follows from the assumptions in Section 4.2. Meanwhile, each of the D-section, skin-stringer panels and ribs are modelled as simply supported plates, with shell edge loads on one side. The skin-stringer panels in Fig. 19 can be split, via the ribs, into 4 simply-supported composite plates. An example setup is seen in Fig. 20.

5.2 FEA Results

Once critical buckling loads are obtained from Abaqus, they are used to normalise the maximum applied direct or shear load to obtain a buckling score. Comparison of the maximum analytical and computational buckling scores are seen in Table 6. The rib designs are verified as the FEA results align very well with the analytical results, as seen in Table 6. Although FEA suggests the D-section does not buckle, there is a large discrepancy with the analytical value. This suggests that the quasi-isotropic method used in Section 4.3 is not suitable for a conservative composite D-section design. Another reason that should also be explored is the varying curvature of the shell, whereas the ESDU 02.03.18 assumes a constant curvature [18]. This could be remedied by splitting the D-section into an upper

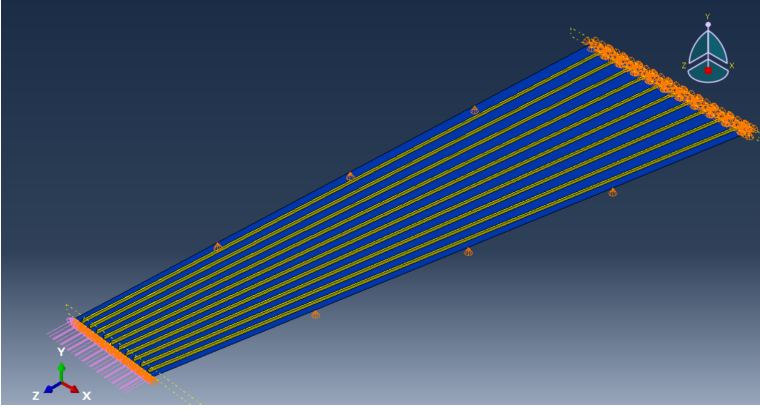


Figure 20. Skin stringer panel at 1.15 - 2.3 m FEA setup, yellow is stringers, blue is skin

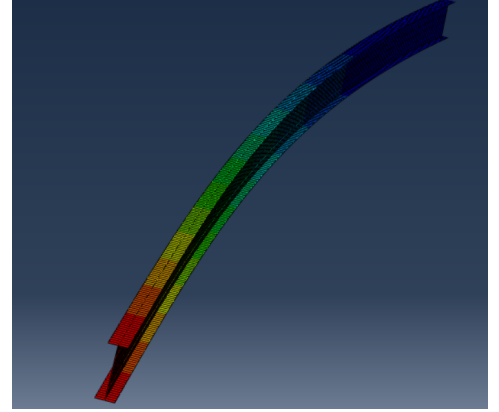


Figure 21. I-beam spar, first buckling mode

and lower curved plate and performing analysis on those instead. The FEA results also differ greatly for the spar under direct stress, with Table 6 suggesting the spar will buckle via the web in a global mode seen in Fig. 21 and corresponds to a buckling score of 6.31. This suggests the assumption made in Section 4.2 where flange and web resist only direct and shear stress respectively is invalid. The focus of future design iterations for the spar should focus on ensuring the web can also resist buckling due to direct stresses. Due to mass constraints, a sandwich structure should be considered for the spar web, with the lightweight core providing improved flexural rigidity while minimising mass.

Table 6. FEA buckling comparison with analytical

Component	Buckle Type	Max Computational Buckling Score	Max Analytical Buckling Score
Spar	Direct	6.31	1.00
Spar	Shear	0.04	0.08
D-Section	Shear	0.65	0.09
Skin-Panel (0-1.15 m)	Direct	N/A	0.10
Skin-Panel (1.15 - 2.3 m)	Direct	N/A	0.06
Skin-Panel (2.3 - 2.7 m)	Direct	0.35	0.91
Skin-Panel (2.7 - 3.0 m)	Direct	0.03	0.14
Ribs	Direct	0.50	0.47

Interesting results are obtained for the skin-stringer panels. The two innermost skin-stringer panels have a first buckling mode shown in Fig. 22. This is unexpected, with first mode of buckling in skin-stringer panels usually looking like Fig. 23. This is due to the 250,000 element limit of the Abaqus Student License preventing the formation of a high fidelity model that can fully capture the skin-stringer interaction of the designed skin panels, as discussed in Section 4.4.1. Meanwhile, for the outermost skin panels with no stringers, the first buckling mode is shown in Fig. 24. Here, FEA yields much lower buckling scores in Table 6, suggesting the analytical critical buckling load is greatly underestimated here. This suggests that approximating the trapezoidal panels as rectangle plates is too conservative, and considering the plates as trapezoids rather than rectangles could allow stringers to be terminated earlier, thus saving further wing mass.

6 Discussion and Next Steps

An all-composite wing with each wing weighing 7.47 kg is proposed in the first design iteration, thus achieving the 7.5 kg design constraint. Both analytical and computational techniques suggest

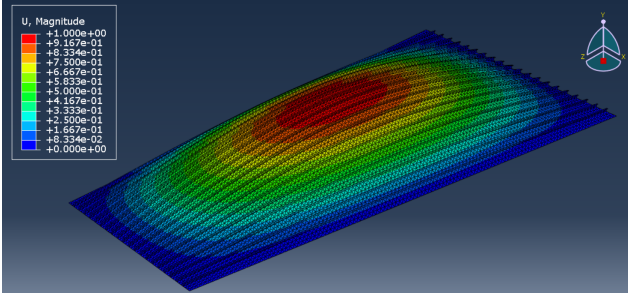


Figure 22. Skin stringer panel at 0-1.15 m, first buckling mode

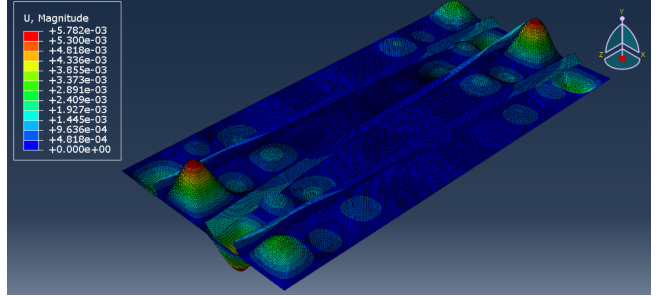


Figure 23. Example simply supported skin-stringer panel, first buckling mode

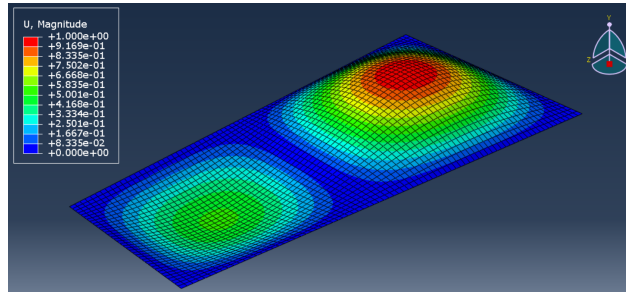


Figure 24. Skin panel at 2.3-2.7 m, first buckling mode

the proposed ribs, outer skin panels and D-section design satisfy buckling and strength constraints. However, FEA suggests the analytical sizing methods for D-section and skin-stringer panels could be improved, as they yielded greatly different results. Using isotropic techniques to size the quasi-isotropic composite layup did not seem to yield accurate results, although using constant curvature datasheets for the varying curvature panel may also be at fault. D-Section FEA was not computationally expensive, so this component could be sized via FEA in future iterations. Meanwhile, treating the skin panel as a rectangle rather than trapezoid was shown to be too conservative in FEA. In future iterations, treating skin panels as trapezoids will ensure the critical buckling load is not greatly underestimated, allowing for stringers to be removed at an earlier span station, thus further reducing mass. This reduction is necessary to accommodate an increase in spar mass as FEA suggests the spar web will buckle. This was due to the assumption of the flange and web only taking direct and shear stress respectively, whereas in reality both experience direct and shear stress. This will be considered analytically in the next design iteration. Although FEA was unable to verify the inner skin-stringer panel designs, models proposed in [19] could be used in future for modelling and verification.

Meanwhile, it was noted that the skin is 77% of the wing mass. Spar placement originally focused on minimising spar mass by considering spar shear flow, but future spar placement should instead focus on minimising skin mass by minimising the compressive force experienced by the skin due to bending moment. This will also involve trade-off studies with Aerodynamics, as thicker airfoils will reduce bending stress and wing mass but can also reduce aerodynamic efficiency. The wingbox idealisation will also be refined in future iterations. Currently, this simplification neither accounts for skin curvature nor the difference in height between front and rear spar. Aeroelastic effects have also been neglected in this first iteration. This is an important effect, particularly for front swept spars which can cause aerodynamic instability. Moreover, the aeroelastic effects caused by wing-boom-rotor-tail interactions have not been considered, and will likely require significant strengthening of the wing at the boom. Although the mass target is achieved for this design iteration, the mass will likely go up in future iterations once the aforementioned effects are accounted for. In future iterations, entire wing models can be run in FEA to capture the interaction between components, as currently each component is assumed to act independently. This should only be done once the aforementioned recommendations have been implemented.

References

- [1] Hasan Rashid. HALO REACH: Project coordination through systems engineering, risk analysis and conceptual design. GDP report 02020313, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [2] Tharm Chursuk. HALO REACH: Optimising weight and balance through comprehensive powerplant sizing for vtol fixed-wing uav. GDP report 02048996, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [3] Michael Chun-Yung Niu. *Airframe Structural Design*. Hong Kong Conmilit Press Ltd., Hong Kong, 2nd edition, 1988. ISBN 962-7128-09-0.
- [4] Matthew M Homa. HALO REACH: Wing design and computational propeller analysis. GDP report 02038403, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [5] Ruairidh Scott-Brown. HALO REACH: Empennage and lateral control design. GDP report 01734080, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [6] Naboth Dirirsa. HALO REACH: Design and optimisation of winglets & aerodynamic tailplane enhancements. GDP report 02026211, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [7] Max F Lawlor. HALO REACH: Drag analysis methodology and validity. GDP report 02026199, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [8] George Halim. HALO REACH: Propulsion system design. GDP report 02059571, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [9] Jiaxun Leng. HALO REACH: V-n diagram, tail, rotor and propeller loads and structural design. GDP report 02015060, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [10] Zhiyan Wang. HALO REACH: Material, manufacturing, undercarriage, boom and dynamics. GDP report 01911209, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [11] Toray. 3960 prepreg system data sheet. <https://www.toraycma.com/wp-content/uploads/3960-Prepreg-System.pdf>. [Accessed 8th June].
- [12] Lorenzo Iannucci. AERO60002 - Design for carbon fibre composites. https://bb.imperial.ac.uk/ultra/courses/_37002_1/cl/outline, 2024. Lecture slides.
- [13] AT Nettles. Basic mechanics of laminated composite plates. Technical Report 1351, NASA Marshall Flight Centre, Alabama, United States of America, 1994.
- [14] Thickness selection for the flanges and web of a composite I-section beam subject to bending and shear. Technical Report 03013, Engineering Science Data Unit (ESDU), 2003.
- [15] Stephen Tsai & Edward Wu. A general theory of strength for anisotropic materials. Technical Report AFML-TR-71-12, Air Force Materials Laboratory, Ohio, United States of America, 1972.
- [16] Dassault Systemes. Tsai-wu failure criterion. [https://help.solidworks.com/2021/English/SolidWorks/cworks/r_tsaiwu_failure_criterion.htm#:~:text=The%20Tsai%20Wu%20criterion%20is,dilatation%20energy\)%20for%20predicting%20failure](https://help.solidworks.com/2021/English/SolidWorks/cworks/r_tsaiwu_failure_criterion.htm#:~:text=The%20Tsai%20Wu%20criterion%20is,dilatation%20energy)%20for%20predicting%20failure). [Accessed 1st June 2024].
- [17] HyperSizer. Effective laminate elastic properties. https://hypersizer.com/help_7.0/Content/Materials/EffLam/el-properties-stiffness.php. [Accessed 3rd June 2024].

- [18] Buckling stress coefficients for curved plates in shear (axial length exceeding circumferential length, edges simply-supported). Technical Report 02.03.18, Engineering Science Data Unit (ESDU), 1952.
- [19] Enzo Cosentino. *Analysis and design of composite panels with stringer run-outs*. PhD thesis, University of Bristol, 2010.
- [20] Local buckling of compression panels with flanged stringers. Technical Report 71014, Engineering Science Data Unit (ESDU), 1971.
- [21] Lorenzo Iannucci & Daqing Yang. AERO60002 - Airframe design: Loading action on the airframe components. https://bb.imperial.ac.uk/ultra/courses/_37002_1/c1/outline, 2024. Lecture slides.
- [22] Buckling of rectangular specially orthotropic plates. Technical Report 80023, Engineering Science Data Unit (ESDU), 1980.

Appendix

A Load Diagrams

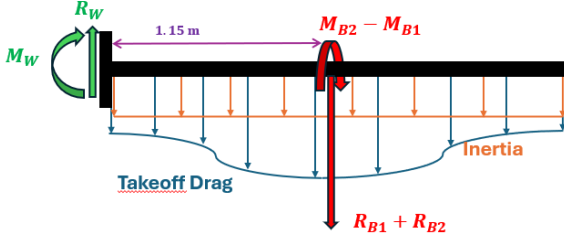


Figure 25. Beam load diagram for takeoff

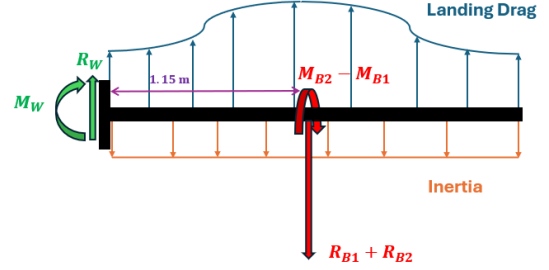


Figure 26. Beam load diagram for landing

Fig. 25 and Fig. 26 denote the load diagrams for takeoff and landing respectively. Note that torque contributions from the boom are omitted from the diagrams for clarity. Similarly, the torque of the wing transmitted to the fuselage is omitted from the diagrams for clarity.

B Load Analysis Subcases

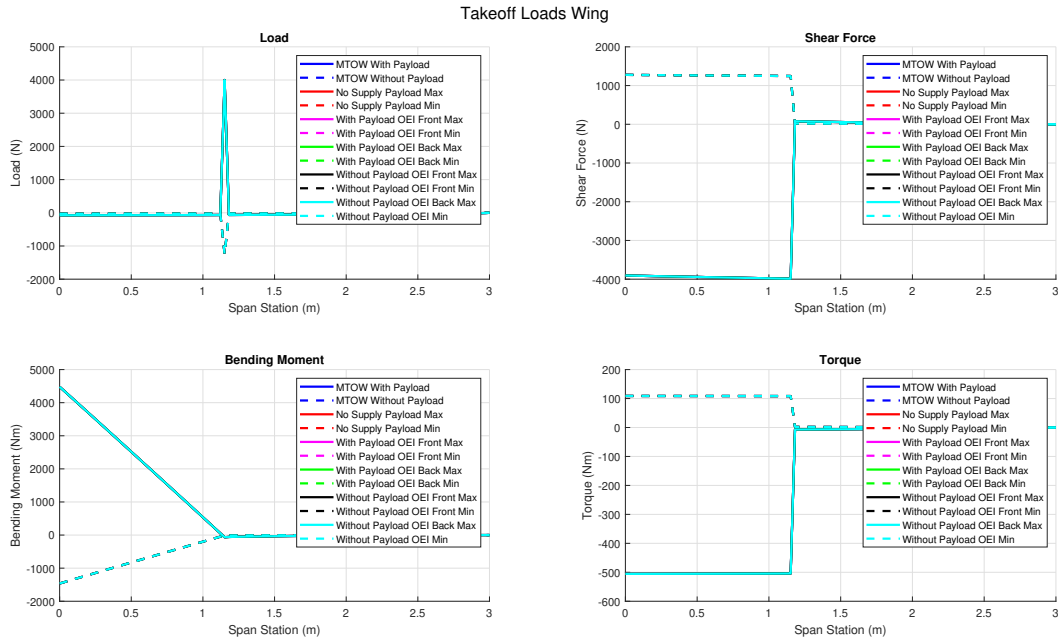


Figure 27. Takeoff loading subcases

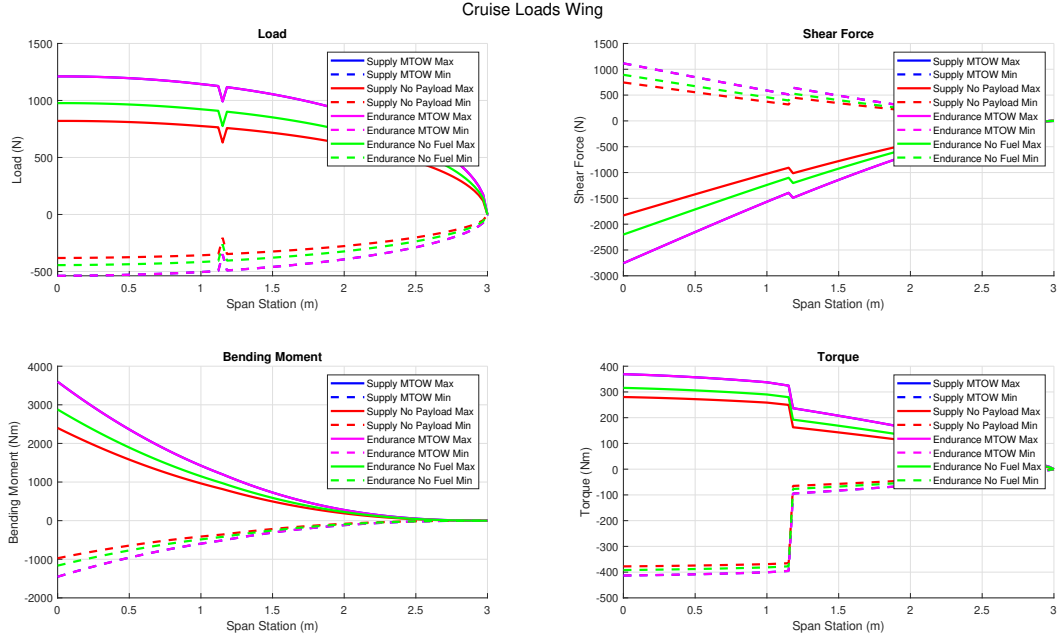


Figure 28. Cruise loading subcases

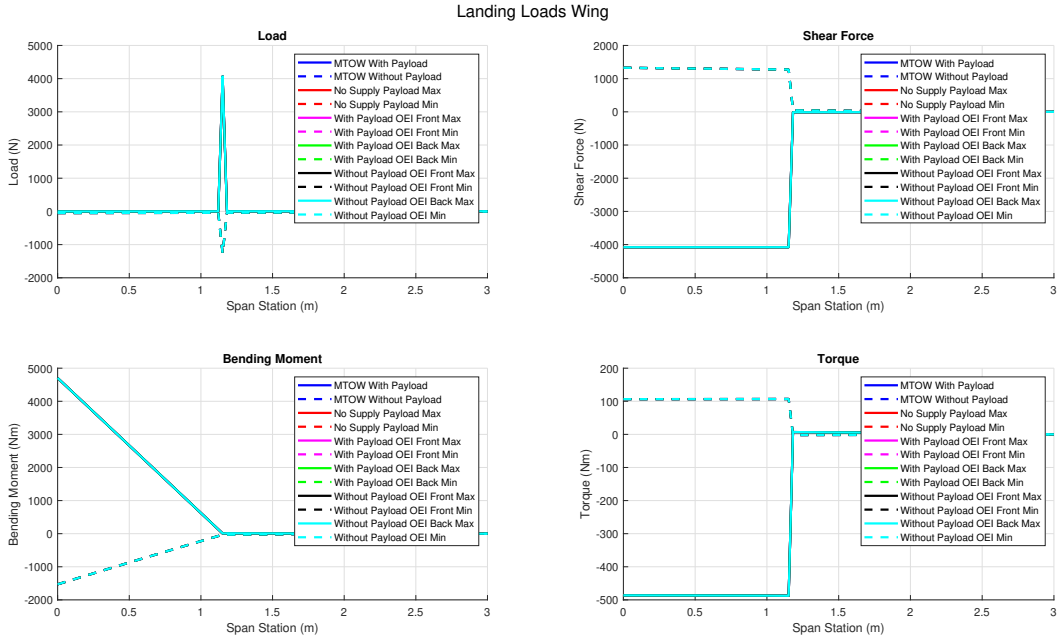


Figure 29. Landing loading subcases

Fig. 27, Fig.28 and Fig. 29 denote the subcases considered for takeoff, cruise and landing respectively. Each case is evaluated at the maximum and minimum load factors, and the most constraining cases are taken and used to plot Fig. 3 - 6.

C Derivation

An I-beam spar can be characterised by four key dimensions: the flange width b_{fl} , flange thickness t_{fl} , web height h_w and web thickness t_w . b_{fl} is a design variable, h_w is fixed as box height and the thicknesses can be obtained upon designing the web and flange layups. Assuming that the bending moment is carried equally by both spars, the stress σ on the flange can be computed using beam theory:

$$\sigma = \frac{My}{2I} \quad (18)$$

where M is the bending moment, I is the moment of inertia and y is the distance from the neutral axis. For an I-beam, the moment of inertia can be computed as:

$$I = \frac{t_w h_w^3}{12} + 2 \left(\frac{b_{fl} t_{fl}^3}{12} + b_{fl} t_{fl} \left(\frac{h_w}{2} \right)^2 \right) \quad (19)$$

As $t_w \ll h_w$ and $t_{fl} \ll b_{fl}$, higher order thickness terms in Eq. (19) are neglected. Note that y is half the web height. Also note that load per unit thickness is desired, which can be computed using $N = \sigma t_{fl}$. Combining this with Eq. (18) and Eq. (19) and simplifying gives Eq. (7).

D FEA

A comprehensive description of FEA model setup, mesh convergence studies and results are presented here to provide more clarity to the FEA investigation performed in Section 5.

D.1 Spars

The spars are modelled as shells according to the dimensions presented in Section 4.2. The composite layup for web and flange, stated in Section 4.2, are implemented via Abaqus composite layups function. One end of the spar has an encastre boundary condition imposed, while the other end is only constrained in the out-of-plane direction. Buckling is investigated by applying a unit shell edge load in a buckling step, with the resulting eigenvalue denoting critical buckling load per unit length. To investigate direct buckling, a normal load is applied to one side of the flange, as seen in Fig. 30; to investigate shear buckling, a shear load is applied to the web. S4R shell elements are used to mesh the part. Mesh convergence studies can be found in Fig. 32 and Fig. 33.

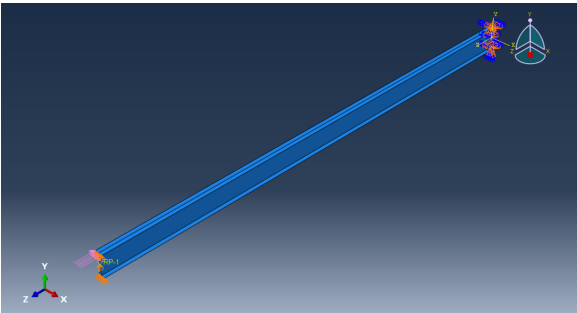


Figure 30. Spar direct buckling model setup

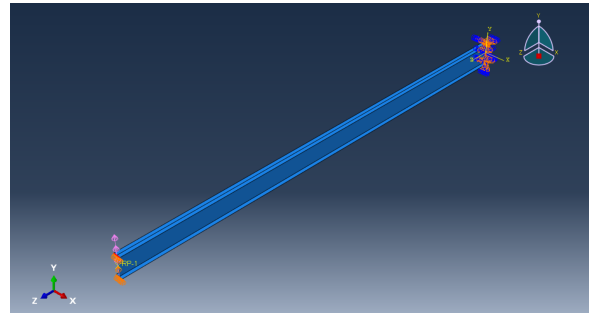


Figure 31. Spar shear buckling model setup

The direct buckling critical load is 2.7×10^5 N/m and the shear critical load is 9.2×10^5 N/m. The maximum direct applied load per unit length is 1.68×10^6 N/m (obtained via Eq. (7)), giving a direct buckling score of 6.31. The maximum shear flow experienced is obtained from Fig. 10 and is 3.6×10^4 , giving shear buckling score of 0.04. These values are recorded in Table 6. The first buckling modes of spar under direct and shear buckling can be seen in Fig. 34 and Fig. 35.

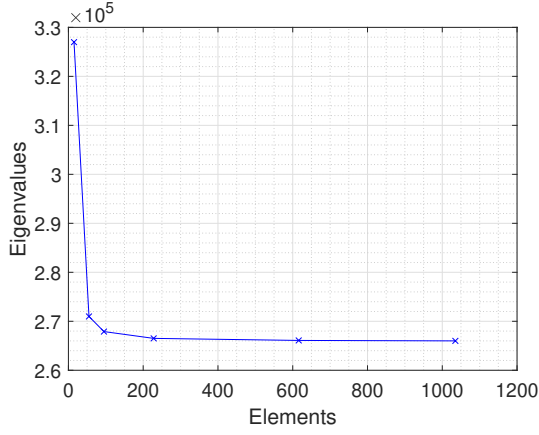


Figure 32. Spar direct buckling mesh convergence study

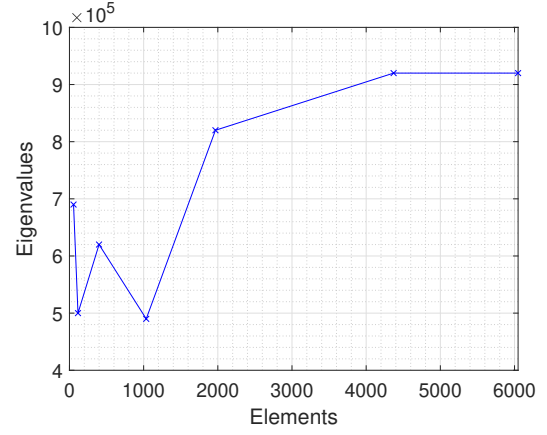


Figure 33. Spar shear buckling mesh convergence study

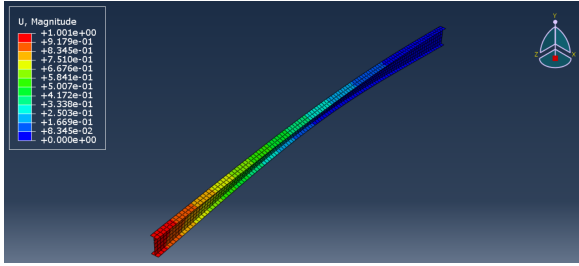


Figure 34. First buckling mode of spar under direct load on flange

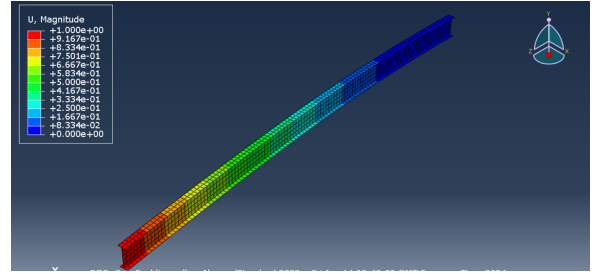


Figure 35. First buckling mode of spar under shear load on web

D.2 D-Section

The D-Section is modelled as shells according to the dimensions presented in Section 4.3. The composite layup, stated in Section 4.3, are implemented via Abaqus composite layups function. The D-section is modelled as a simply supported curved plate. Shear buckling is investigated by applying a unit shear shell edge load in a buckling step, with the resulting eigenvalue denoting critical buckling load per unit length. Model setup can be seen in Fig. 36. S4R shell elements are used to mesh the part. Mesh convergence study can be found in Fig. 37.

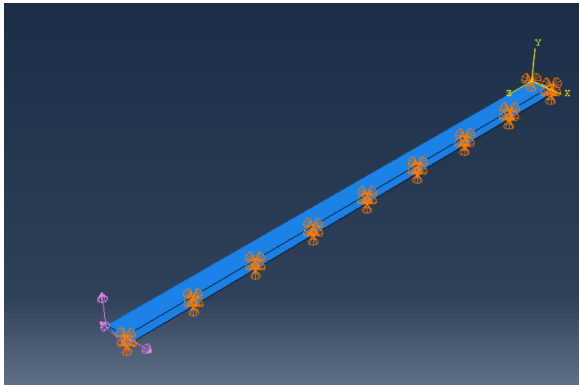


Figure 36. D-Section FEA model setup

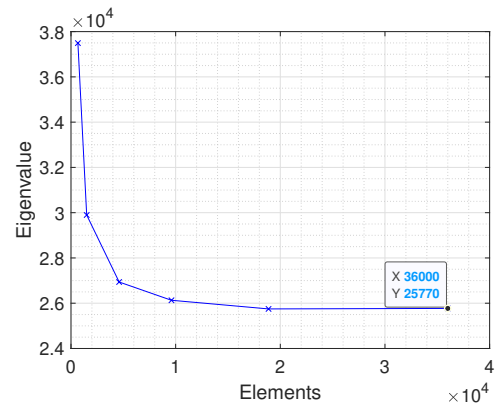


Figure 37. D-section mesh convergence study

The critical shear buckling load is 2.6×10^4 N/m and the maximum applied shear flow is 1.7×10^4 in Fig. 10. This gives a buckling score of 0.65, as seen in Table 6. The first buckling mode can be

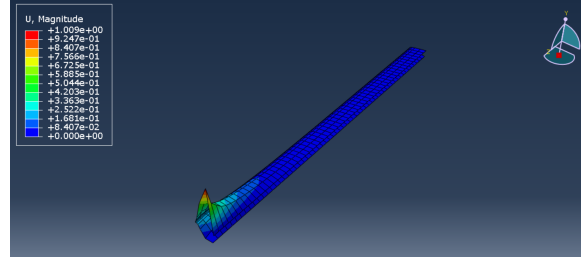


Figure 38. D-section first shear buckling mode

seen in Fig. 38.

D.3 Skin-Stringer Panels

The two innermost skin-stringer panels (0-1.15 m and 1.15 - 2.3 m) are modelled as shells according to the dimensions in Section 4.4.3. The composite layup, stated in Section 4.4.3, are implemented via Abaqus composite layups function. Simply supported boundary conditions are imposed on all edges, and buckling is investigated by applying a unit shell edge load in a buckling step, with the resulting eigenvalue denoting critical buckling load per unit length. To investigate buckling, a normal load is applied to the outer edge, with model setup mentioned in Section 5. S4R shell elements are used to mesh the part. No mesh convergence study is presented here as neither meshes converged before the 250,000 element limit of the Abaqus Student License was reached. This prevented the accurate capture of skin-stringer interaction, as mentioned in Section 5. No buckling score is derived, and FEA first buckling modes have already been mentioned and discussed in Section 5.

D.4 Skin Panels

The two outer skin panels (2.3 - 2.7 m and 2.7 - 3 m) are modelled as shells according to the dimensions in Section 4.4.3. The composite layup, stated in Section 4.4.3, are implemented via Abaqus composite layups function. Simply supported boundary conditions are imposed on all edges, and buckling is investigated by applying a unit shell edge load in a buckling step, with the resulting eigenvalue denoting critical buckling load per unit length. To investigate buckling, a normal load is applied to the outer edge, with model setup same as skin-stringer panels but without the stringers. S4R shell elements are used to mesh the part. Mesh convergence studies are found in Fig. 39 and Fig. 40.

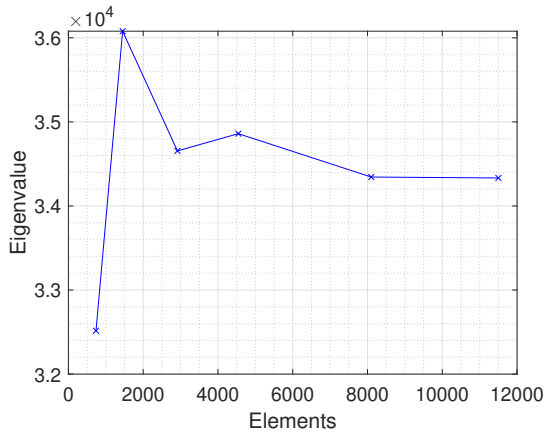


Figure 39. Skin panel at 2.3-2.7 m mesh convergence study

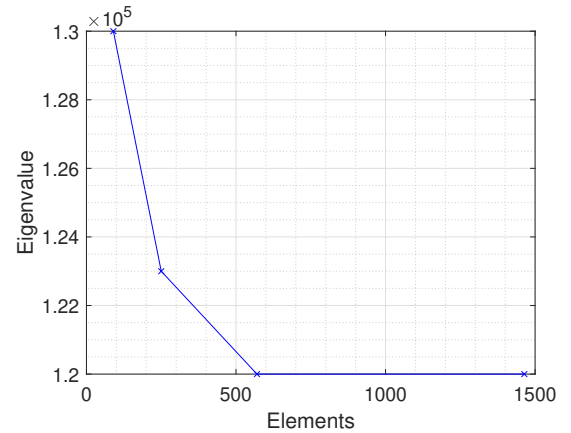


Figure 40. Skin panel at 2.7-3.0 m mesh convergence study

For the skin panel at 2.3 - 2.7 m, the maximum applied load experienced is 1.2×10^4 N/m (acquired

by dividing the bending moment at 2.3 m by the wingbox length and height at 2.3 m). The critical buckling load is 3.4×10^4 N/m, yielding a buckle score of 0.35. For the skin panel at 2.7 - 3.0 m, the maximum applied load experienced is 4.1×10^3 N/m and the critical buckling load is 1.2×10^5 N/m, yielding a buckle score of 0.03. The buckle scores are seen in Table 6. First buckling modes are already presented in Section 5.

D.5 Ribs

The ribs are modelled as shells according to the dimensions presented in Section 4.2. The composite layup, stated in Section 4.4.3, are implemented via Abaqus composite layups function. A low aspect ratio simply supported plate is modelled and buckling is investigated by applying a unit shell edge load in a buckling step, with the resulting eigenvalue denoting critical buckling load per unit length. Model setup is seen in Fig. 41. Note only rib at root is modelled as this is the most constraining case with greatest crushing force. S4R shell elements are used to mesh the part. Mesh convergence studies can be found in Fig. 42.

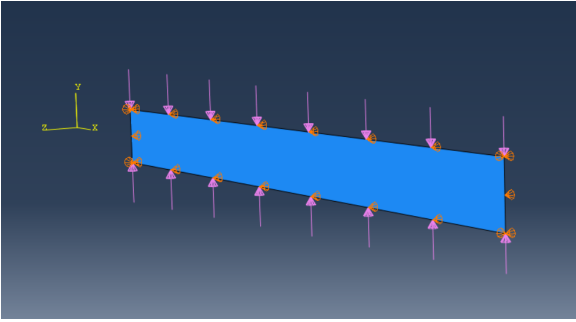


Figure 41. Rib at root FEA model

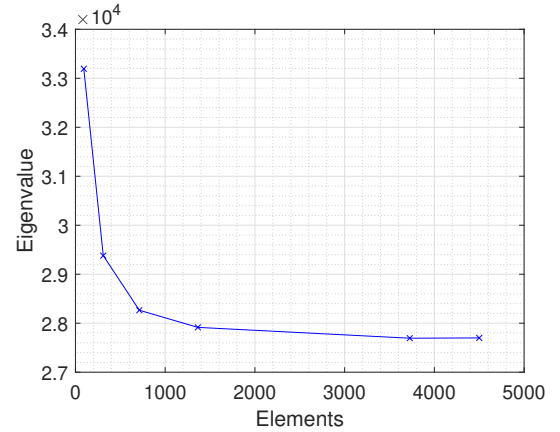


Figure 42. Rib at root mesh convergence

The max crushing force experienced at the root is 8,889 N, yielding a load of 1.4×10^4 N/m acting on the ribs. The rib critical buckling load is 2.8×10^4 , yielding a buckle score of 0.50, as seen in Table 6. The first buckling mode is seen in Fig. 43.

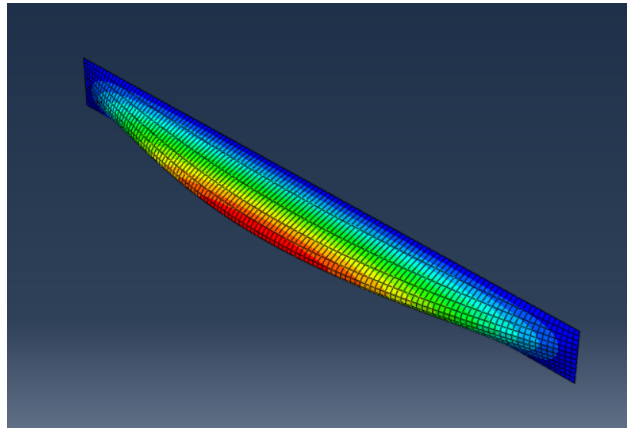


Figure 43. Rib at root, first buckling mode